

B22: Space Jam

Discord name	Affiliation	Experience	Role
Kishor R	National Institute of Technology Karnataka	Undergraduate	Team lead
Rohit L	National Institute of Technology Karnataka	Undergraduate	Member
Rishadd Ranjith	National Institute of Technology Karnataka	Undergraduate	Member

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Goal

The primary objective is to design and implement a reliable, space grade mechanical fault detector as a standalone digital ASIC on the GlobalFoundries 180nm platform by executing Goertzel algorithm locally to isolate structural and bearing anomalies without the overhead of continuous raw data streaming.

Design

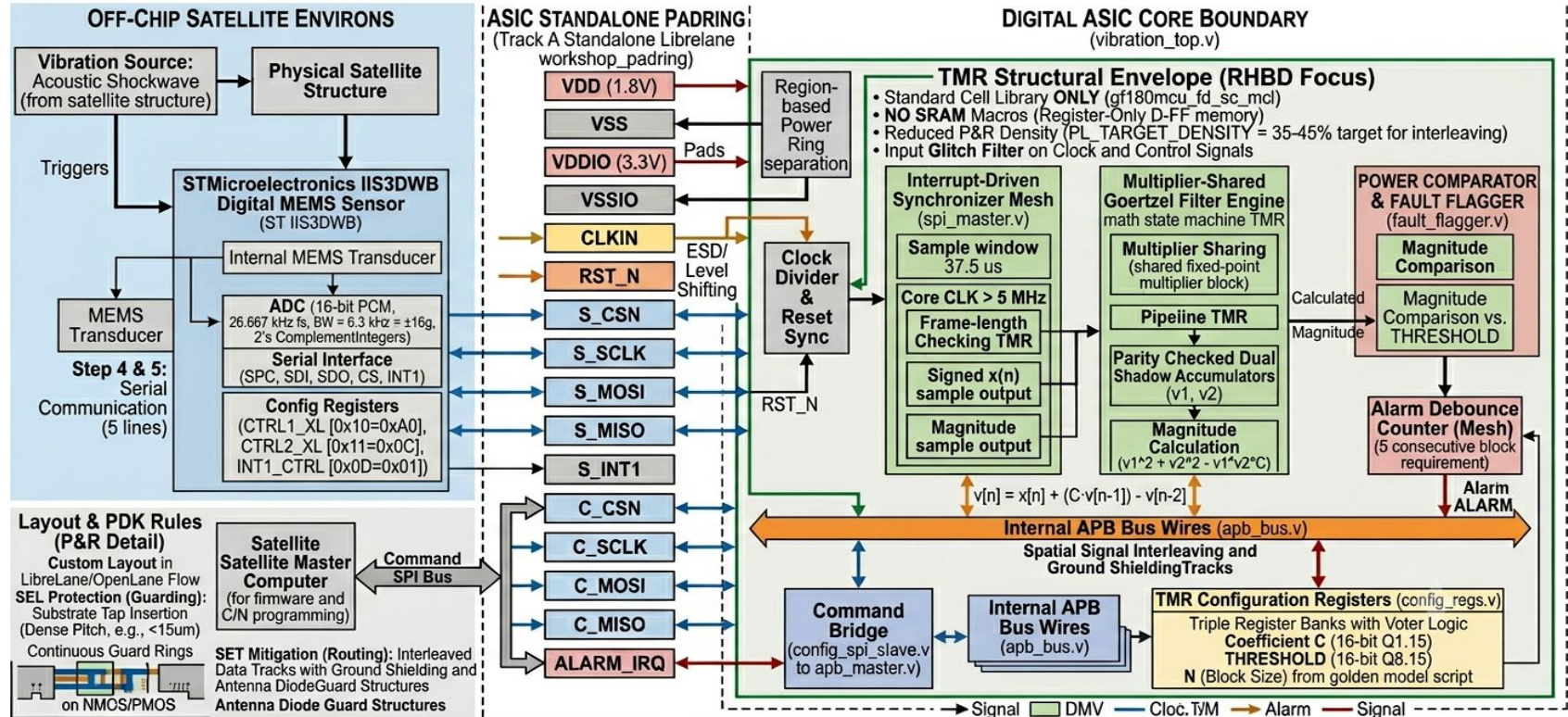
- interfaces with an off-chip STMicroelectronics IIS3DWB digital MEMS vibration sensor via a dedicated SPI master to process 16-bit PCM acceleration samples natively at 26.667 kHz
- incorporates an internal Advanced Peripheral Bus (APB) interface driving dynamically programmable configuration registers hardened with Triple Modular Redundancy {Radiation hardening by Design}
- Other radiation protection schemes:
 - Enclosing guard rings
 - Delay based filtering mesh for signals (for signals like reset)
 - Increased antenna diode insertion
 - Dense tap cell placement

Repository:

<https://github.com/Kishor5115/Space-Grade-Mechanical-Fault-Detector>

note : image generated using AI Prompt

Vibrational Sensor Based Space-Grade Mechanical Fault Detector



Application

This ASIC acts as an autonomous edge-computing anomaly watchdog for monitoring the structural health and bearing wear of satellite reaction wheels and other deployment mechanisms in deep space environments.

Resources:

[Datasheet - IIS3DWB - Ultrawide bandwidth, low-noise, 3-axis digital vibration sensor](https://github.com/sscs-ose/sscs-chipathon-2026/tree/main/examples/librelane_rtl2gds_gf180)
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Team background

- Academic Experience

- All three of us are undergraduate students at NITK India
- Have dabbled into VLSI and really focused on to Digital IP Design

- Work Experience

- Kishor R: Intern at I2S Technologies Pvt. Ltd. specialising in Addressable LEDs.
- Rohit L: Research Intern at IISc (CSRL Lab) specialising in FIA on PQC schemes on MCUs
- Rishadd Ranjith: Intern at Intel India specialising in PCIe

Questions, Suggestions, Doubts?

